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Deep Multi-sphere Support Vector Data Description

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Abstract

Deep learning is increasingly used for unsupervised feature extraction and anomaly detection in big datasets. Most deep learning based anomaly detection techniques separately train a neural network for feature extraction, then apply a traditional anomaly detection method on the extracted features. These hybrid techniques have achieved higher accuracy than traditional anomaly detection methods and reconstruction-error-based deep autoencoders. However, recent research demonstrates that jointly optimising the objectives of the deep network and the anomaly detection technique in a hybrid architecture substantially improves detection performance. Existing methods that use this objective assume that the normal (i.e., non-anomalous) data comes from a single distribution. In this paper, we show that violation of this assumption negatively affects performance of these methods and creates model bias in the favour of anomalies. We propose Deep Multi-sphere Support Vector Data Description, which jointly optimises the objectives of the deep network and anomaly detection. It generates useful and discriminative features by embeding normal data with a multi-modal distribution into multiple data-enclosing hyper-spheres with minimum volume. We empirically show that our proposed method outperforms state-of-the-art shallow and deep anomaly detection methods.

1 Introduction

Anomaly detection (AD), also known as outlier detection, discovers rare instances in data that differ from the patterns observed in the majority of the instances [1]. AD is applied in different domains ranging from intrusion detection [2, 3] to health and safety related applications such as disease diagnosis and fall detection for elderly people [4, 5]. The common aspect of all these applications is detecting unexpected observations, which are often translated to events that negatively affect the monitored system [1]. A key challenge for AD is the scarcity of labelled data [6]. The high cost of labelling hinders the application of supervised AD techniques, which require labelled examples of anomalies and *normal (i.e., non-anomalous)* data [1]. In some applications however, it is possible to collect the data during the normal operation of the system, which leads to modelling normal instances as a single class and rejecting (as anomalies) future observations that do not conform to this model [1]. This is known as *One-Class Classification* (OCC). Techniques such as *One-Class Support Vector Machine* (OCSVM) [7] and *Support Vector Data Description* (SVDD) [8] are widely used for OCC-based AD.

The performance of traditional AD methods drops for high-dimensional data (e.g., fall detection using video signals) due to the curse of data dimensionality [9]. Except for a few AD methods such as *Isolation Forest* (iForest) [10], training on high-dimensional datasets is also computationally expensive. Feature engineering can compensate for this problem by generating a smaller set of useful features, but manual extraction of these features is also expensive due to the need for a domain expert [11]. Thus, feature extraction techniques have been proposed to automatically extract useful features for high-dimensional datasets [2, 3]. Among feature extraction techniques, deep learning based methods are shown to be powerful in generating useful features for anomaly detection [9].

Early neural network based AD techniques [12, 13] used the normal data to train an autoencoder. Since the reconstruction error of the autoencoder on instances that differ from the training set is likely to be high, this error was used to measure the extent to which an instance was anomalous. Subsequent research [9, 14, 15] proposed hybrid methods that used an autoencoder to extract a small set of useful features as input to traditional AD techniques. These methods train the autoencoder and the traditional AD method sequentially and separately, which lacks feedback from the AD method to the feature extractor. As a result, the features are optimised for the objective of the autoencoder and are not explicitly customised for learning good features for the AD method. Recently, it was shown that jointly optimising the objectives of the deep neural network and the AD technique in a hybrid architecture substantially improves the performance [16–18]. Ruff et al. [16] proposed *Deep SVDD* (DSVDD), and Chalapathy et al. proposed *Deep*

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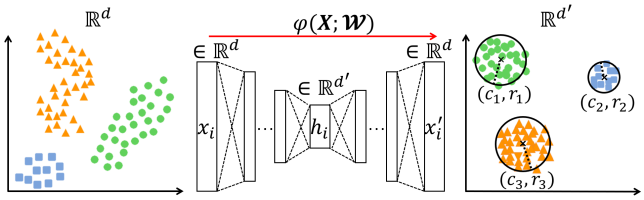

Figure 1: Illustration of the parametric mapping $\phi(.;\mathcal{W})$ to transform $X \subseteq \mathbb{R}^d$ to $\phi(X;\mathcal{W}) \subseteq \mathbb{R}^{d'}$. The three arbitrary-shaped clusters in X are mapped to $K = 3$ hyper-spheres in $\mathbb{R}^{d'}$ described by their center c_k and radius r_k for $k = 1..K$.

OCSVM (DOCSVM) [17] as Deep OCC methods, which jointly optimise the objective of an autoencoder and an OCC based AD method. DSVDD is based on the objective function of the SVDD algorithm [8], while DOCSVM is based on OCSVM [7]. These techniques achieved better performance compared to other hybrid as well as shallow techniques [7, 16]. However, they assume that normal data comprises a single cluster (i.e., a unimodal distribution). This assumption is not valid for many real-life applications as several different underlying processes can generate the normal data.

In this paper, we propose *Deep Multi-sphere SVDD* (DMSVDD), which addresses this challenge by training a neural network while minimising the average volume of multiple data-enclosing hyper-spheres in the network representation of the data (Figure 1). DMSVDD creates a compact representation of the different underlying distributions in the data by transferring them to hyper-spherical clusters in the feature space. This is different from both DSVDD [16], which embeds the data into a single hyper-sphere, and DOCSVM [17], which separates the feature vectors from the origin using a maximum-margin hyper-plane. In addition to making no assumptions regarding the underlying distribution of the normal data, DMSVDD makes no assumption regarding the number of clusters. Instead, it starts from a large value for this parameter and automatically adjusts it during the optimisation process.

We show that DMSVDD provides a statistically significant improvement in the accuracy compared to the state-of-the-art related techniques, especially when there are multiple clusters in the normal data. In addition, we show that the network representation of the data, i.e., the extracted features, generated using DMSVDD better preserve the structure of the normal data and its distribution, as opposed to its single-sphere counterpart DSVDD. Consequently, this improved representation enhances the explainability of the model. Our experiments demonstrate that DSVDD is biased towards assigning the highest anomaly scores to specific types of anomalies, which can negatively affect its performance when the top ranked anomalies are investigated in an exploratory analysis [1]. Unlike DSVDD, the top ranked anomalies scored by DMSVDD comprise different types of anomalies, which means DMSVDD trains models with lower bias.

Our work is different from [19] and [20], which generate synthetic counter examples of anomalies in the input or feature space to improve the network data representation as well as AD accuracy using a discriminative instead of generative learning approach. Wen et al. [21] proposed a discriminative deep feature learning approach for face recognition to improve the accuracy of multi-class classification. This supervised approach jointly minimises two losses. The softmax loss, *computed using the ground truth labels*, aims to maximise inter-class separation. The second loss, which minimises the intra-class variation, is computed by taking the average of the distances between each instance and its class centroid. This is different from the objective of DMSVDD, because our method is unsupervised, meaning that there are no ground truth labels to specify the true number of classes and update the classes' centroid given the true class labels. For similar reasons, our method is different from [22], a supervised multi-class classification method that uses SVDD layers in a neural network to avoid overfilling.

2 Problem Definition and Notation

This section explains the notation used in this paper and provides a formal definition of the problem. Let $\mathcal{X} \subset \mathbb{R}^d$ comprise all normal (i.e., non-anomalous) data instances in the input space $\mathbb{R}^d$. Data in $\mathcal{X}$ comes from an unknown distribution that can be unimodal or multimodal. Therefore, data points in $\mathcal{X}$ form the clusters $C = \{c_1, ..c_k\}$, where $k = 1..K$ and the value of K is unknown. Clusters in the data have arbitrary sizes and shapes in the input space $\mathbb{R}^d$. The aim of anomaly detection is to define a scoring function $f(x)$ such that:

$$f(x) < f(\tilde{x}); \forall x \in \mathcal{X}, \tilde{x} \notin \mathcal{X}. \tag{2.1}$$

Only a subset of $\mathcal{X}$, denoted by $\mathbf{X} = \{x_1, ..., x_n\}$, is available as a training set for learning $f(x)$. In other words, examples of $\tilde{x} \notin \mathcal{X}$ are not available. Thus, finding a scoring function $f^*(x)$ that does not violate (2.1) is not feasible. Instead, the scoring function $f(x)$ should be estimated using only the data instances in $\mathbf{X}$.

3 Background and Related Work

In this section, we briefly review SVDD and the literature on multi-sphere SVDD. We also provide a short survey of the existing hybrid Deep AD techniques that jointly optimise the objectives of the neural network and the OCC-based AD method.

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3.1 SVDD: Tax and Duin [8] proposed SVDD to minimise the radius R of a data-enclosing hyper-sphere in a *Reproducing Kernel Hilbert Space* (RKHS) associated with the kernel function $\phi(.,\theta)$:

(3.2)
$$\min_{R,a,\xi} R^2 + \frac{1}{\nu n}\sum_{i=1}^{l}\xi_i, \;\; s.t. \; \|\phi(x_i,\theta)-c\|^2 \leq R^2+\xi_i,$$

where $\xi_i \geq 0$ for $i = 1..n$, c is the center of the hyper-sphere, R is its radius, and θ is the mapping hyper-parameter. A maximum fraction of $\nu \in (0,1)$ data points in X are allowed to fall outside the hyper-sphere by being assigned non-zero slack variables ξ_i.

3.2 Multi-sphere SVDD: SVDD with multiple hyper-spheres [23–25] was proposed for data with a multi-modal distribution. Similar to Hao and Lin [23], Le et al. [25] proposed a new optimisation problem for SVDD by substituting the radius R and the center c with a vector of radius values and a matrix of hyper-sphere centers. They proposed an iterative algorithm that alternates between finding the hyper-sphere centers, their radius and the slack variables, and assigning data points to the hyper-spheres. Le et al. [26] also proposed a fuzzy multi-sphere SVDD, which uses fuzzy membership for the hyper-spheres. Xiao et al. [24] assumed that $\phi(X;\theta)$ is explicitly available. They argued that the feature space generated using the kernel function may comprise multiple hyper-spheres. To address this issue, they first consider each data point in X to be a data enclosing hyper-sphere. For each x_i, the number of data points n_i in a predefined distance from x_i is counted. Then, x_i is considered as a center if n_i is greater than a relatively small predefined value. After this initialisation, procedures for growing and merging the hyper-spheres are repeated to respectively increase the radius of hyper-spheres or merge them together until a stopping criterion is reached. Despite the difficulty of setting the values of the predefined parameters, this technique requires accessing the values of $\phi(x_i,\theta)$, which are not available in the case of widely used non-linear kernels such as the RBF kernel.

3.3 Deep hybrid OCC with AD Objective: Ruff et al. [16] proposed the DSVDD method, which uses an SVDD-based objective function for jointly learning the parameters of a hybrid model. The model is a deep encoder followed by an SVDD model for AD. First, an autoencoder is pretrained in an unsupervised manner by minimising the reconstruction error of the training data X. Then, the decoder part is removed and the weights of the encoder network are fine-tuned to minimise the volume of the hyper-sphere that encloses a majority of the network representations of the data. The authors assumed that the normal data comes from a single (unimodal) distribution. Chalapathy et al. [17] made the same assumption and proposed a similar technique, except they used the objective function of the hyper-plane OCSVM for the fine tuning step. Unlike these techniques, our proposed method addresses the challenge of modeling a multimodal (instead of unimodal) distribution of normal data.

4 Deep Multi-sphere SVDD

To estimate $f(x)$ using the data instances in $\mathbf{X}$, a parametric mapping $\phi(.;\mathcal{W}) : \mathbb{R}^d \rightarrow \mathbb{R}^{d'}$ is computed to project the data into a feature space $\mathbb{R}^{d'}$, where $d' \ll d$. The objective is to map clusters of data from the input space into data-enclosing hyper-spherical clusters with minimal volume in the feature space as visualised in Figure 1. By iteratively minimising the volume of the hyper-spheres in the feature space, the hyper-parameter $\mathcal{W}$ is updated such that $\phi(.;\mathcal{W})$ captures common sources of variation in the clusters, i.e., variations created by rare noisy inputs are ignored. Therefore, anomalous data points are highly likely projected outside the data-enclosing hyper-spheres.

Observation 1 (Network-representation-mean-bias). *Let $\phi(.;\mathcal{W}_i)$ be a parametric mapping to $\mathbb{R}^{d'}$ at iteration i of optimising $\phi(.;\mathcal{W})$, and $\phi(X;\mathcal{W}_i)$ comprises more than one cluster. A_j denotes a subset of anomalous observations that will be seen in $R^{d'}$ during the test phase. Let $\overline{A_j} \approx \overline{\phi(X;\mathcal{W}_i)}$, where $\overline{*}$ denotes the mean value of the set. If A_j resides in the interspace of the clusters in $\phi(X;\mathcal{W}_i)$, any update $\phi(.;\mathcal{W}'_{i+1})$ that encloses the data in a hyper-sphere centered at $\overline{\phi(X;\mathcal{W}_i)}$ is biased towards A_j.*

Observation 1 means that if some anomalies lie in the interspace of the clusters in the normal data near to the mean of the data, merging the clusters towards the data mean results in creating a bias in the model. This bias results in assigning low anomaly scores to these types of anomalies as visualised in Figure 2. Anomalies of type A_1 that lie near to the mean value of the data in $\mathbb{R}^{d'}$, will be assigned lower anomaly scores than anomalies of type A_2 when using $\phi(.;\mathcal{W}'_{i+1})$. However, since $\phi(.;\mathcal{W}_{i+1})$ preserves the structure of the clusters in X, it is not affected by this phenomenon. Thus, we propose that $\phi(.;\mathcal{W})$ should preserve the structure of the data to avoid creating bias towards specific types of anomalies, and minimise the following objective function for this aim:

$$\frac{1}{n}\sum_{i=1}^{n}\|\phi(x_i;\mathcal{W})-c_j\|, \tag{4.3}$$

where $\mathbf{C} = \{c_1, ..., c_K\}$ denote the centers of the hyper-

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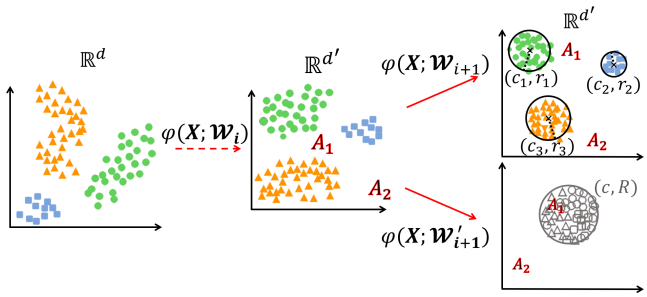

Figure 2: Network-representation-mean-bias: the left figure depicts an arbitrary dataset in $\mathbb{R}^d$ with three clusters. The clusters are visualised for the ease of explanation, but labels are not used in training the models. The middle figure shows an intermediary stage of optimising $\phi(.; \mathcal{W})$, in which the three clusters are well-separated. Let A_1 and A_2 denote the location of the test set anomalies in $\mathbb{R}^{d'}$. The bottom-right figure demonstrates an update in $\phi(.; \mathcal{W})$ that merges all the clusters around the data mean in $\mathbb{R}^{d'}$, whereas the upper-right figure shows an update that preserves the data structure. When data is merged into one cluster using $\phi(.; \mathcal{W}'_{i+1})$, the model assigns low scores to instances from A_1 as they are now colocated with normal data points in c.

spherical clusters in $\mathbb{R}^{d'}$, $\|.\|$ is the L^2 (Euclidean) norm, and c_j is the cluster center assigned to $\phi(x_i; \mathcal{W})$ using the *nearest neighbour prototype rule* (NPR), i.e., $j = \underset{k}{argmin} \|\phi(x_i; \mathcal{W}) - c_k\|$. DMSVDD uses a two phase optimisation explained in the next section (Algorithm 1) to solve (4.3).

4.1 Methodology. We choose $\phi(.; \mathcal{W})$ to be a neural network with L number of layers. Thus, $\mathcal{W} = \{\mathbf{W}_1, ..., \mathbf{W}_L\}$ is a set of weight matrices, which are jointly learned to minimise the average volumes of K data-enclosing hyper-spheres in the feature space $\mathbb{R}^{d'}$, characterised by centers $\mathbf{C} = \{c_1, ..., c_K\}$ and radius vector $r = [r_1, .., r_K]$:

(4.4)
$$\underset{\mathcal{W}, r}{min} \frac{1}{K} \sum_{k=1}^{K} r_k^2 + \frac{1}{\nu n} \sum_{i=1}^{n} max(0, \|\phi(x_i; \mathcal{W}) - c_j\|^2 - r_j^2) + \frac{\lambda}{2} \sum_{l=1}^{L} \|W^l\|,$$

where c_j is the cluster center assigned to $\phi(x_j; \mathcal{W})$ using NPR and $r_j > 0$ is its corresponding radius. The first term in (4.4) minimises the volume of hyper-spheres while the second term penalises for mapping points outside of the hyper-spheres and is controlled by parameter $\nu \in (0, 1)$. This parameter allows a fraction of points to be mapped outside the hyper-spheres to compensate for noisy instances or unknown anomalies in the training data X. The last term is a network

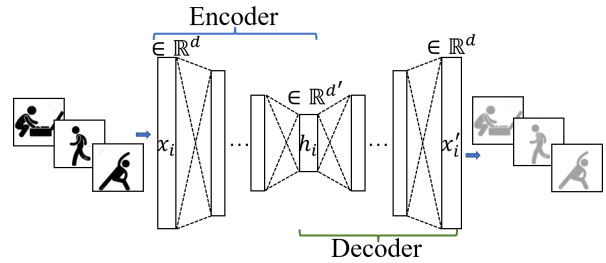

Figure 3: Initialisation of $\mathcal{W}$ using an autoencoder. After a pretraining phase using the autoencoder, the encoder part is used for transforming data points $x_i \in X$ from the input space $\mathbb{R}^d$ to $h_i = \phi(x_i, \mathcal{W}) \in \mathbb{R}^{d'}$.

weight decay regulariser on $\mathcal{W}$ with parameter $\lambda > 0$.

To initialise $\mathcal{W}$, a symmetric autoencoder is used as depicted in Figure 3. The autoencoder minimises the reconstruction error of the training set, i.e., $\frac{1}{n} \sum_{i=1}^{n} \|x_i - x'_i\|$. After initialisation of the network, X is mapped to $H = \phi(X; \mathcal{W}) = \{h_1, .., h_n\} \subset \mathbb{R}^{d'}$ (line 5 in Algorithm 1). The K-means algorithm [27] is used to choose K cluster centers $\mathbf{C} = \{c_1, ..., c_K\}$ in H. The cluster membership $U \in \mathbb{R}^{K \times n}$ is computed using NPR to identify crisp partitions in H. Cardinality of cluster center c_k is $n_k = \sum_{i=1}^{n} u_{ki}$, and $u_{ki} \in \{0, 1\} = 1$ where $k = \underset{j=1..K}{argmin} \|h_i - c_j\|$. The value of K is initialised using a large value K_{init} (line 4 in Algorithm 1), and adjusted automatically (line 19 in Algorithm 1) during the optimisation process.

To solve (4.4) by computing $\mathcal{W}$ and r, an alternating optimisation procedure is used after the initialisation phase. In the first step of the optimisation (lines 8 to 15 in Algorithm 1), the value of r_k for cluster c_k is computed as follows:

- If the cluster cardinality is less than ν times the maximum cardinality of the clusters (i.e., $n_k < \nu \times max([n_1, .., n_K])$), then $r_k = 0$. This implies that the cluster cardinality is not large enough to be considered as a separate cluster, and members of this cluster are either noise/anomalies or belong to a bigger neighboring cluster.

- Otherwise, r_k is assigned with the value of the $(1 - \nu)^{th}$ quantile of distances computed for the members of c_k. This means $\nu \times n_k$ members of c_k can to fall outside its corresponding hyper-sphere.

The second step of the alternating optimisation procedure fixes the value of r and computes the new value of the weight set $\mathcal{W}$ by the standard backpropagation of the errors computed using (4.4) in the network. For this step, we use Adam [28], which is a gradient-based stochastic optimization technique, in min-batch mode. *Absence of members of a cluster in a mini-batch does not cause any problem due to*

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Algorithm 1 Deep Multi-sphere SVDD (DMSVDD)

Inputs: $X \subset \mathbb{R}^d$, ν, K_{init}
Output: $\phi(.;\mathcal{W})$, C, r, K
initialise $\phi(.;\mathcal{W})$ using an autoencoder ▷ Phase 1
$K = K_{init}$
$H = \phi(X;\mathcal{W})$ ▷ Map $X \subset \mathbb{R}^d$ to $H \subset \mathbb{R}^{d'}$ using $\phi(.;\mathcal{W})$
$(C, U) = Kmeans(H, K)$ ▷ Initialise cluster centers in H using K-means [27]
while convergence is not achieved **do** ▷ Phase 2: Optimisation loop
  for k = 1 **to** K **do** ▷ Step 1: initialise radius vector r
    $n_k = \sum_{i=1}^{n} u_{ki}$ ▷ $u_{ki} \in \{0,1\} = 1$ where $k = \underset{j=1..K}{argmin} \|h_i - c_j\|$
    if $n_k < \nu \times max([n_1, .., n_K])$ **then**
      $r_k = 0$
    else
      $r_k = (1-\nu)$th quantile of the distances $\|h_i - c_k\|$ where $u_{ki} = 1$
    end if
  end for
  compute $J(\mathcal{W}, C, r)$ using Equation (4.4)
  Adam(J,$\mathcal{W}$) ▷ Step 2: update $\mathcal{W}$ for 4.4 using Adam optimiser [28]
  $H = \phi(X;\mathcal{W})$
  $K = \sum_{k=1}^{K} 1_{\{r_k > 0\}}$
  delete cluster centers with $r_k = 0$ from C
end while

the taken alternating optimisation approach: during the optimisation of $\mathcal{W}$, the cluster set C and the radius vector r are not modified; if a cluster is missing in a mini-batch, its corresponding feature vectors are not updated after processing that mini-batch.

After the convergence, the anomaly score of a test data point x is computed as $s(x) = \|\phi(x;\mathcal{W}) - c_k\|^2 - r_k^2$, where $x \in c_k$ by NPR.

5 Empirical Evaluation

This section evaluates our proposed DMSVDD method with shallow and deep anomaly detection methods[1]. SVDD [8] and Kernel Density Estimation (KDE) with the *Radial Basis Function* (RBF) kernel, and iForest [10] are used as shallow comparison techniques. For deep architectures, DSVDD [16] and *Deep Convolutional Autoencoders* (DCAE) [29] are used. The reconstruction error of DCAE is used as the anomaly score. Since OCSVM [7] and SVDD with the RBF kernel converge to similar solutions, we report the results for SVDD. For a similar reason, DOCSVM [17] is excluded from the analysis. The *Area Under the Curve* (AUC) for the *Receiver Operating Characteristic* (ROC) curve is used to measure the accuracy of anomaly detection techniques [16]. For deep techniques, we also compare the network representations of the data using t-SNE [30].

5.1 Datasets. We use the MNIST [31], CIFAR10 [32], and the second version of the MobiAct [33] datasets. All the datasets are scaled in the range $[0, 1]$ using min-max-scaling based on the training set. The MobiAct dataset comprises eleven daily activities such as walking, sitting, jumping, etc., and four types of falling on the floor, which are anomalies. There are 67 subjects including male and female participants, from which we chose a subset that have performed all the activities. Since we are using a convolutional autoencoder, a preprocessing step should be applied on this time series data to make it ready for being processed by the convolutional layers. For this reason and in keeping with the human activity recognition literature [34,35], we used a fixed length sliding-window of 2 seconds on the time-series data from the (x, y, z) axis of the accelerometer with 50% overlap. If a fall activity exists in a window, the corresponding data instance is labelled as an anomaly. For the training set, we also used sampling to balance the data generated for each class of activities. It comprised $10,000$ instances all from the normal activities. The test set has $5,000$ instances of normal activities and fall chosen at random from the remaining data.

For the MNIST dataset, we change the normal concept to create different datasets. In MNIST0, digit 0 is selected as normal and the rest of the digits are considered as anomalies. This dataset provides a basis for comparing DMSVDD and DSVDD [16] when normal data comes from a single distribution. In MNIST01, digits 0 and 1 are considered as normal and the rest of the digits are considered as anomalies. These two digits have different distributions, while digits $\{2, 3, 5, 6, 8, 9\}$ have similar distributions to digit 0, and digits $\{4, 7\}$ have distributions similar to digit 1. Therefore, this experiment identifies how accurate the models are when having two clusters of normal data. To make the AD task more difficult, the MNIST013 dataset considers digits $\{0, 1, 3\}$ as normal and the rest of the digits as anomalies. In this case, digits $\{2, 5, 6, 8, 9\}$ have

[1] https://github.com/zghafoori/Deep-Multi-Sphere-SVDD

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Table 1: The accuracy of DMSVDD for the examined datasets in comparison with other methods.

Dataset	SVDD	KDE	IF	DCAE	DSVDD	DMSVDD
MNIST0	0.945 ±0.001	0.675 ±0.006	0.981 ±0.003	**0.995** ±0.000	**0.994** ±0.000	**0.994** ±0.001
MNIST01	0.517 ±0.002	0.832 ±0.004	0.911 ±0.007	0.992 ±0.001	0.984 ±0.001	**0.997** ±0.000
MNIST013	0.626 ±0.002	0.730 ±0.006	0.835 ±0.007	0.969 ±0.002	0.923 ±0.015	**0.976** ±0.019
MobiAct	0.503 ±0.003	0.837 ±0.010	0.777 ±0.012	0.933 ±0.006	0.849 ±0.055	**0.974** ±0.006
CIFAR-10	**0.697** ±0.002	0.581 ±0.006	0.579 ±0.010	0.602 ±0.020	0.664 ±0.021	0.663 ±0.022
Average	0.658 ±0.002	0.731 ±0.006	0.817 ±0.008	0.898 ±0.006	0.883 ±0.019	**0.921** ±0.010

distributions similar to both digits 0 and 3, which means they lie in the interspace of these two classes. The aim is to study Observation 1 (Network-representation-mean-bias) for DSVDD compared to our DMSVDD. Training was performed on the normal instances from the training set of the MNIST dataset and testing was performed using both normal and anomalous instances in its test set.

For the CIFAR10 dataset, vehicles, i.e., automobiles and trucks, were considered as the normal data and classes that represent animals were treated as anomalies. The deep convolutional autoencoder is unable to generate a good initial data representation that differentiates among different types of vehicles for this data. Thus, we use it to show that DMSVDD and DSVDD converge to a similar solution if the initialisation of the data representation using the autoencoder is not trustworthy. Training was performed on the normal instances from the training set of the CIFAR10 dataset and testing was performed using both normal and anomalous instances in its test set.

5.2 Experimental Setup. We used the implementation provided by Ruff et al. [16] in their Deep SVDD toolbox[2] for the comparison techniques. The default settings in the toolbox were used [16]. For ease of reference, we summarise the default settings here. PCA was used to reduce the data dimensionality for shallow methods by choosing principal components that preserve 95% of the variance in the data. A fraction of 10% randomly selected test instances was used for grid-search to set the RBF kernel width in SVDD from the set $\{2^{-10}, 2^{-9}, .., 2^{-1}\}$. This gives a small supervised advantage to SVDD for setting its hyperparameters. Following Ruff et al. [16], ν is changed in the set $\{0.01, 0.05, 0.1\}$ and the best result is reported. The bandwidth of the Gaussian kernel in KDE is searched in the set $\{2^{0.5}, 2^{1}, .., 2^{5}\}$ via 5-fold cross-validation using the log-likelihood score. The default setting of iForest [10], i.e., 100 trees and a sample size of 256 instances, was used. *For our DMSVDD method, $K_{init} = 100$ is used as the initial number of clusters for all the datasets, and it is left to DMSVDD to reduce this number to the true number of separable clusters in the normal data using the mechanism explained in Algorithm 1.* In our study, we changed this parameter from 10 to 100 with a step-size of 10 and the results were unchanged, therefore we report the results for $K_{init} = 100$.

For the deep networks, the Adam optimizer [28] is used with the parameters recommended in the original work. Network weight initialisation is performed using uniform Glorot weights [36]. We used Batch Normalization [37] as suggested by Ruff et al. [16] to normalise the output of the intermediate layers of the deep networks. The Leaky ReLU activation is used with leakiness $\alpha = 0.1$. All the deep networks are pretrained for 150 epochs and then another 100 epochs of fine-tuning based on their corresponding objective function. The batch size is set to 200, and the weight decay parameter $\lambda = 10^{-6}$. For all the datasets, we used the LeNet type CAEs similar to Ruff et al. [16].

5.3 Results and Discussion. We ran each experiment 10 times with different seeds to take into consideration the random initialisations in different methods. Table 1 shows the average AUC of 10 runs for each dataset and method along with the variation of the results over the 10 runs. Except for CIFAR-10, the deep techniques perform better than the shallow methods. We discuss each dataset in the following.

For the MNIST0 dataset, with unimodal distribution for normal data, DCAE, DSVDD and DMSVDD have similar accuracy. This is because the convolutional autoencoder is able to separate instances of digit 0 from the other digits and generates a good initial data representation. DMSVDD converges to a single cluster, by automatically setting the radii of 99 cluster centers out of $K_{init} = 100$ to zero.

Adding another non-anomalous cluster into MNIST01 results in a slight decrease in the accuracy of DCAE and DSVDD and considerable negative effect on the accuracy of the shallow techniques (Table 1). On the other hand, the accuracy of DMSVDD remains high for this dataset. Figure 4(a) visualises the network representation of the deep approaches on a randomly chosen subset of the *test set* for MNIST01 using t-SNE [30]. The visualisation for DCAE (Figure 4(a)-DCAE) is provided as a baseline to show the improvement achieved by DMSVDD

[2]https://github.com/lukasruff/Deep-SVDD

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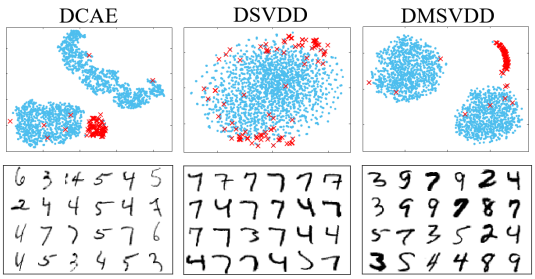

Figure 4: Visualisation of the MNIST01 test set: (first row) the network representation H for DCAE, DSVDD and our DMSVDD respectively from left to right, and (second row) the top ranked anomalies scored by each of the techniques in the same order (blue points: normal data; red crosses: anomalies). DMSVDD identifies the two clusters in the data and generates two hyper-spherical clusters in H. Unlike DSVDD, the top ranked anomalies in DMSVDD are from different types of anomalies. DSVDD has a bias towards assigning higher anomaly scores to digits 4 and 7.

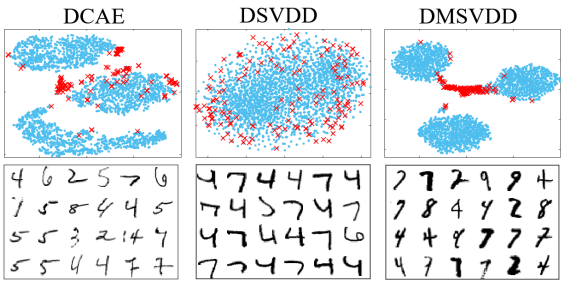

Figure 5: Visualisation of the MNIST013 test set: (first row) the network representation H for DCAE, DSVDD and our DMSVDD respectively from left to right, and (second row) the top ranked anomalies scored by each of the techniques in the same order (blue points: normal data; red crosses: anomalies). DMSVDD identifies and generates three hyper-spherical clusters in H. Unlike DSVDD, the top ranked anomalies in DMSVDD are from different types of anomalies. DSVDD has a bias towards assigning higher anomaly scores to digits 4 and 7.

(Figure 4(a)-DMSVDD), i.e., the two arbitrary shaped non-anomalous clusters in DCAE compared to the well-shaped hyper-spherical ones in DMSVDD. With $K_{init} = 100$, DMSVDD has successfully detected two clusters in the normal data and separated them from each other as well as from the anomalies. Therefore, it has the best accuracy for MNIST01. In contrast, as Figure 4(a)-DSVDD shows, DSVDD merges the two clusters of digits 0 and 1, which increases the chance of embedding anomalies into the enclosing hyper-sphere and decreasing the detection performance of this method. To investigate Observation 1 (Network-representation-mean-bias) for the MNIST01 dataset, Figure 4(b) depicts the top ranked anomalies for DCAE, DSVDD and DMSVDD. It shows that DSVDD tends to favour anomalous instances of digits $\{2, 3, 5, 6, 8, 9\}$, which have distributions similar to digit 0. Therefore, the top ranked anomalies in DSVDD are mainly from digits 4 and 7. In scenarios like intrusion detection, in which the top ranked anomalies are investigated first, this problem makes the system highly vulnerable to the favoured anomalies. Figure 4(b)-DMSVDD shows that our DMSVDD method identifies different types of anomalous instances as top ranked, which reduces its model bias towards a specific type of anomalies.

For MNIST013, Table 1 reports the best result for DMSVDD followed by DCAE. The accuracy of all the shallow methods is considerably decreased for this dataset. The accuracy of DSVDD is decreased considerably because anomalous instances from digits $\{2, 5, 6, 8, 9\}$ lie in the interspace of the normal digits 0 and 3, and reside close to the data mean in $\mathbb{R}^{d'}$ (Observation 1). Thus, many anomalous instances overlap with the normal data in its hyper-spherical data representation in 5(a)-DSVDD. Figure 5(b)-DSVDD confirms Observation 1, as the top ranked anomalies by DSVDD only belong to digits 7 and 4. DMSVDD does not have this problem and correctly identifies the three clusters (Figure 5(a)-DMSVDD), which are well-shaped and separated compared to CDAE (Figure 5(a)-CDAE).

The shallow methods also perform poorly in detecting falls for the MobiAct dataset (Table 1). DMSVDD has the best accuracy followed by CDAE. Figure 6-DCAE shows that most of anomalies are very close to the mean of the network representation of the data. Based on Observation 1, DSVDD is biased towards assigning low scores to such anomalies. Thus, it maps anomalies into the enclosing hyper-sphere (Figure 6-DSVDD) and its performance for this dataset is considerably decreased to be similar to the shallow methods. On the other hand, our DMSVDD method separates the normal data to three clusters and maps most of anomalies outside the boundaries of the enclosing hyper-spheres (Figure 6-DMSVDD).

Finally for the CIFAR-10 dataset, the best accuracy in Table 1 is for SVDD, followed by DSVDD and DMSVDD. However, SVDD has the advantage of using 10% of the test set for setting its parameters, whereas DSVDD and DMSVDD use no ground-truth labels. In this experiment, the network representation of the data after pretraining does not separate the two normal classes, therefore DMSVDD converges to a solution similar to DSVDD by merging all the training data into a single hyper-sphere.

To assess the statistical significance of the differences in the accuracy shown in Table 1, Table 2 shows the results of a Wilcoxon signed-rank test. All the techniques are compared to DMSVDD, which had

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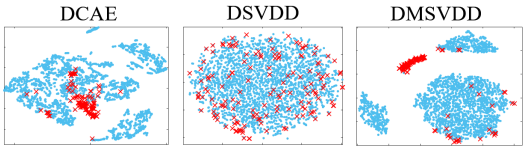

Figure 6: Visualisation of the MobiAct test set: the network representation H for DCAE, DSVDD and our DMSVDD respectively from left to right (blue points: normal data; red crosses: anomalies). DMSVDD identifies three clusters in the data and separates them from anomalies. In contrast, DSVDD embeds anomalies into the enclosing hyper-sphere.

the highest average accuracy over all of the examined datasets in Table 1. The results from 10 runs of the 5 datasets were used for the calculations. This test investigates to what extent the null hypothesis H_0, which indicates no difference between two methods in terms of their accuracy, can be rejected. For each comparison, the test returns the sum of positive ranks (R^+) and negative ranks (R^-) of the first method, and the p-value. The p-value represents the lowest level of significance of a hypothesis that results in a rejection, and if it is less than the significance level α, then the null hypothesis H_0 can be rejected. This means the improvement is significant at the level α. A level of significance of $\alpha = 0.05$ is used for this purpose. Table 2 shows that DMSVDD provides a statistically significant improvement compared to the other techniques as the p-values are all smaller than 0.05.

The shallow methods ran on a CPU and deep methods ran on a GPU. Therefore, similar to [16], we do not report the running times. Among the shallow methods, iForest has the lowest time complexity of $O(t \times \psi \times log\psi)$, where $\psi = 256$ and $t = 100$ in our experiments. The time-complexity of grid search for SVDD and KDE is $O(n^3)$, and for applying PCA for the shallow methods is $O(min(d^3, n^3))$. The time complexity of DCAE per epoch is $O(\sum_{l=1}^{L} p_{l-1} \times s_l^2 \times p_l \times q_l^2)$. In this formula, p_* denotes the number of channels in layer $*$, which is the number of convolutional filters for the hidden layers, and $l = 0$ denotes the input layer of the network, while s and q respectively show the size of the filter and output in the corresponding layer. In our experiments, the pretraining of all the deep methods have the same complexity. DSVDD and DMSVDD have extra computations for the distance to the hyper-sphere centers compared to DCAE, which adds the extra term $O(B \times |C| \times d')$ for their fine tuning step, where $B = 200$ denotes the batch size. In DSVDD, $|C| = 1$ is fixed, thus its extra tuning step time-complexity is $O(B \times d')$. For DMSVDD, $|C|$ starts from $K_{init} = 100$ and converges to the actual number of clusters after a few iterations. Let K denote the true cardinality of C. Since this convergence to K is very fast and happens in the initial epochs, the extra tuning step time-complexity of DMSVDD is $O(B \times K \times d')$, which is always K times bigger than DSVDD. During the test phase, the time complexity of DSVDD per instance is $O(d')$ and for DMSVDD is $O(K \times d')$.

Table 2: Wilcoxon test (DMSVDD vs other methods).

DMSVDD vs.	SVDD	KDE	IF	DCAE	DSVDD
R^+	1199	1275	1275	1063	892
R^-	76	0	0	113	98
p-value	2.9E-11	1.0E-20	1.9E-15	9.1E-07	1.3E-05

6 Conclusions

We proposed DMSVDD, a hybrid anomaly detection technique with an unsupervised objective function. DMSVDD uses a deep neural network followed by a multi-sphere SVDD model to embed the normal data into multiple data-enclosing hyper-spheres with minimum volume. Unlike earlier works that suffer when the normal data has a multi-modal distribution, DMSVDD automatically identifies the number of separable modes of the normal data and embeds them into different hyper-spheres. Our experiments show that this new technique provides a statistically significant improvement over the existing shallow and deep anomaly detection methods and is not biased towards a specific type of anomalies. All existing hybrid methods that jointly optimise the objective of the neural network and SVDD/OCSVM have used the primal formulation of these anomaly detection algorithms. In future, we aim to investigate the use of the dual formulation and kernels such as RBF.

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